

Particle in a 1-D Box

✓ Please Enter your Name, Roll No, Section below:

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Motivation

To demonstrate how to use the Schrodinger equation we will look at one of the simplest quantum problems: particle in a box. This problem simply asks what are the solutions to the Shrodinger equation for a particle that is allowed to travel in one dimension in a finite domain.

Learning Goals

After working through these notes, you will be able to:

1. Solve the Schrodinger equation for a 1D particle in a finite box
2. Plot 1D particle in a box wave functions.

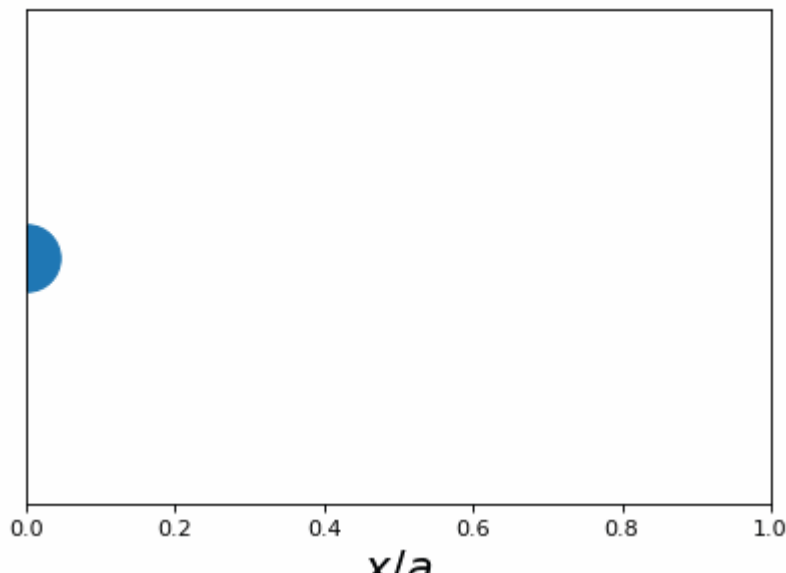
Coding Concepts

The following coding concepts are used in this notebook:

1. [Variables](#)
2. [Functions](#)
3. [Plotting with matplotlib](#)

✓ Particle in a Box: Setting up the Problem

To see the utility of the Schrodinger equation we will see what it predicts for the energy of a particle that can travel in one dimension but is restricted to a finite domain. This can be seen as asking what the allowed energy is for a particle in a one dimensional box. The only energy that this particle has is kinetic energy.



The first step in solving this problem, as is the case for all problem involving the Schrodinger equation, is to write out the Hamiltonian. From there, we can hopefully solve for the wave functions and the energies. To do so we can start by writing out the classical energy and then converting the terms to quantum operators.

$$\begin{aligned} E_{classical}(x, v) &= K(v) + V(x) \\ &= \frac{1}{2}mv^2 \end{aligned}$$

where we have used that $V(x) = 0$ as is stated in the problem. So the total energy is just equivalent to the total kinetic energy. Since this is for a single particle the classical kinetic energy is just $\frac{1}{2}mv^2$.

In order to write out the Hamiltonian we can always start with

$$\begin{aligned} \hat{H} &= \hat{K} + \hat{V} \\ &= \hat{K} \end{aligned}$$

where $\hat{K}$ is the kinetic energy operator. If we consult the table provided in the [Schrodinger Equation notes](#) we see that

$$\hat{K} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right).$$

Since this problem only involves a single dimension, we can write

$$\hat{K} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}.$$

Thus, the Schrodinger equation for this problem is

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E\psi(x) \quad 0 \leq x \leq a$$

and the goal is to determine $\psi(x)$ and E . And the energy eigenvalues are given by:

$$E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

✓ Particle in a Box: Applications

Although this seems like a trivial problem, the model of a free particle in a finite box has been applied to actual physical systems with reasonable success. One example is the π electrons in a conjugated molecule. The free electron model for butadiene does a reasonable job at reproducing the $\pi \rightarrow \pi^*$ absorption band.

✓ Probabilistic Interpretation of the Wave Function

The wave function of a particle, such as the free particle, can be interpreted as the amplitude of the wave at that position. This is analogous to the classical wave equation.

The probability of finding a particle at position x is proportional to the square of the wave function. Specifically,

$$P(x) \propto \psi^*(x)\psi(x),$$

where $\psi^*(x)$ is the complex conjugate of $\psi(x)$. Note that wave functions can be complex but, if they are not, then $\psi^*(x) = \psi(x)$ and $P \propto \psi(x)^2$.

This interpretation of the wave function has some interesting and useful outcomes. Here we will only consider three of them:

1. The wave function must be normalized
2. Probability of observing a particle in a particular domain is

$$\text{Prob}(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} \psi^*(x)\psi(x)dx$$

3. Average properties of a particle can be computed as

$$\langle A \rangle = \int \psi^*(x)\hat{A}\psi(x)dx$$

✓ Wave functions must be normalized

Because the square of the wave function is related to the probability we can normalize the wave function based on

$$\int_{-\infty}^{\infty} P(x)dx = \int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx = 1$$

Let's consider what this means for the particle in a box wave functions.

$$\begin{aligned}\psi_n(x) &= B_n \sin \frac{n\pi x}{a} \\ \psi_n^*(x)\psi_n(x) &= B_n^* B_n \sin^2 \frac{n\pi x}{a}\end{aligned}$$

We can determine the value for B_n based on the normalization condition.

$$\int_0^a \psi_n^*(x) \psi_n(x) dx = |B_n|^2 \int_0^a \sin^2 \frac{n\pi x}{a} dx = 1$$

where we have use the domain $0 \leq x \leq a$ rather than $-\infty < x < \infty$ because that is the finite domain of this particular problem. Now let's solve the $\sin^2$ integral

$$\int_0^a \sin^2 \frac{n\pi x}{a} dx = \frac{a}{2}$$

Thus, we have that

$$B = \sqrt{\frac{2}{a}}$$

Finally, our normalized particle in a box wave functions are

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

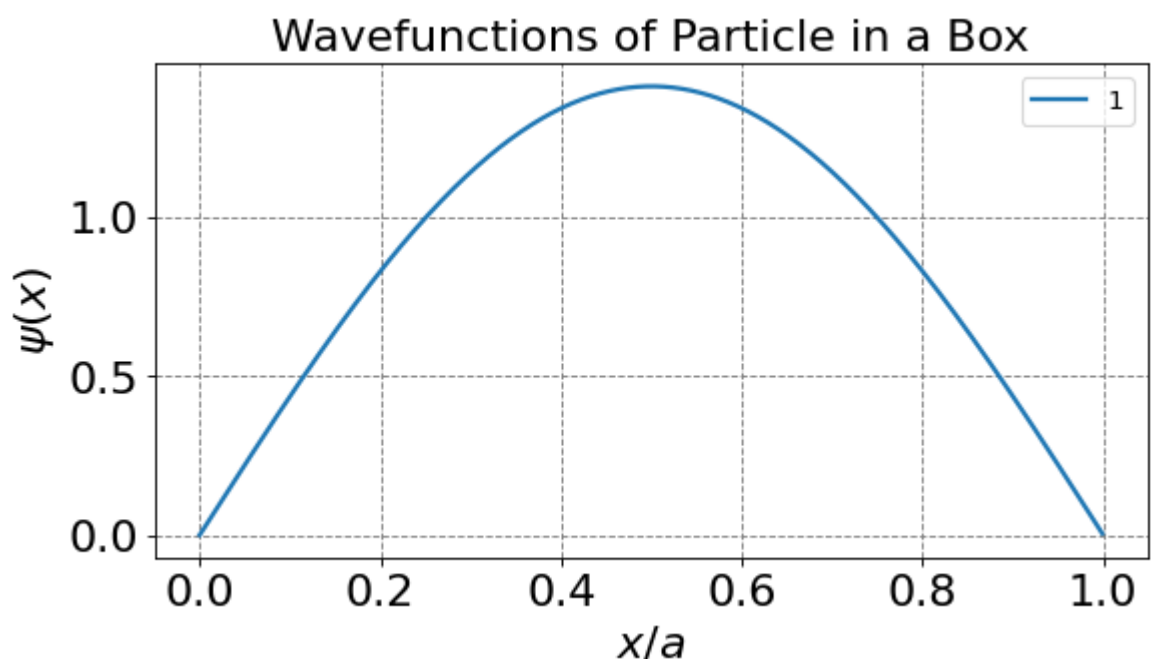
Show code

```
# make an array containing domain of wavelengths to consider
x = np.arange(0,1,0.001) # X is range
```

```
a = 1 # L length of the box
```

```
n = 1 # Energy level
```

Show code



Lab Exercises (Student Handout)

Learning Objectives

By the end of this lab, you should be able to:

1. Visualize how energy levels arise due to confinement.
2. Relate box length to energy spacing.
3. Interpret wavefunctions and probability density.
4. Connect quantum confinement to real materials (nanotechnology).

Exercise 1: Energy Level Pattern Recognition

Run the energy-level calculation for **L = 1 nm** and fill the table:

n	Energy (eV)	Spacing vs previous
1	___	–
2	___	Larger / Smaller
3	___	Larger / Smaller
4	___	Larger / Smaller
5	___	Larger / Smaller

Question: As n increases, spacing between levels:

- ☐ Increases
- ☐ Decreases
- ☐ Remains constant

```
a=1 # L length of the box for Energy Calculation Energy level is same.  
n_val = 5 # Energy level
```

[Show code](#)

n = 5, Energy = 9.4008 eV

Exercise 1: Manual Entry Table

Please fill in the calculated values directly in this table: [Double click the table.]

n	Energy (eV)	Spacing vs previous
1	0.3760	--
2	1.5041	larger
3	3.3843	larger
4	6.0165	larger
5	9.4008	larger

✓ Exercise 2: Box Length vs Energy (What-if)

Change $L = 1 \text{ nm} \rightarrow 2 \text{ nm} \rightarrow 5 \text{ nm}$ and record E_1 :

L (nm)	E_1 (eV)
1	___
2	___
5	___

Concept check: Increasing L makes E_1 :

- ☐ Increase
- ☐ Decrease
- ☐ Stay the same

```
a=5 # L length of the box should be varied.
n_val = 1 # Energy level is constant in this exercise.
```

[Show code](#)

$L = 5$, Energy = 0.0150 eV

Exercise 2: Manual Entry Table

Please fill in the calculated values directly in this table: [Double click the table.]

L	Energy (eV)	Spacing vs previous
1	0.3760	--
2	0.0940	decrease
3	0.0418	decrease
4	0.0235	decrease
5	0.0150	decrease

✓ Exercise 3: Wavefunction Shape & Nodes

Plot $\psi(x)$ for $n = 1, 2, 3$.

Fill:

n	Nodes inside the box
1	0
2	1
3	2

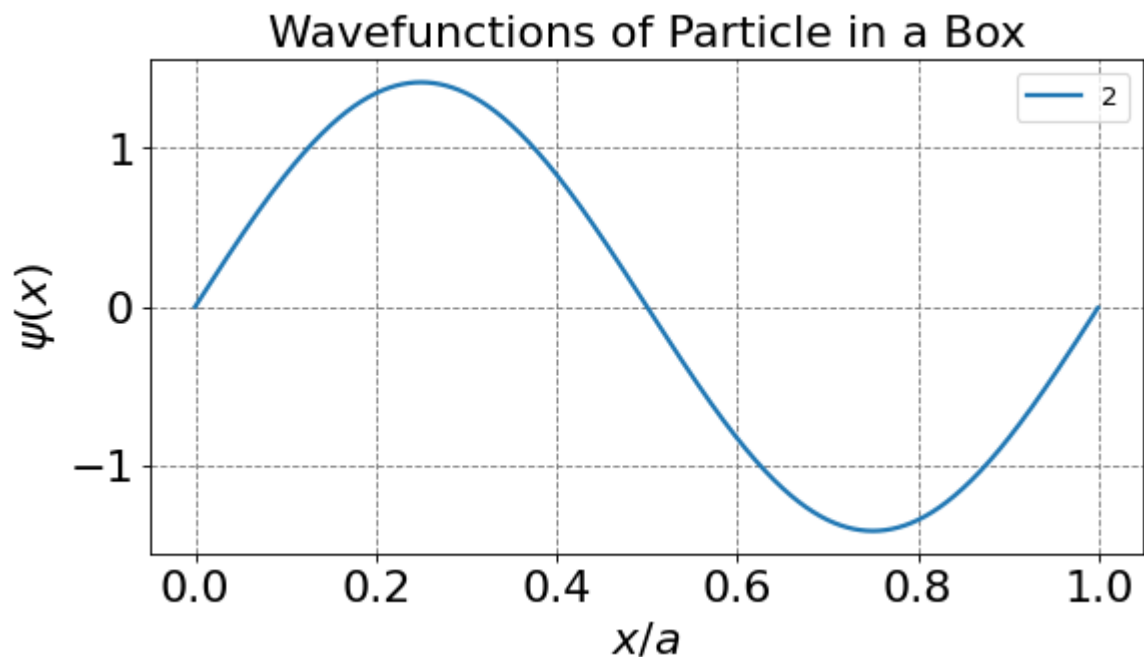
Choose:

- ☐ nodes = n
- ☐ nodes = n - 1
- ☐ nodes = 2n

```
# make an array containing domain of wavelengths to consider
x = np.arange(0,1,0.001) # X is range
```

```
a=1 # L length of the box
n = 2 # Energy level
```

Show code



✓ Exercise 4: Probability Density Insight

Plot

$$|\psi_n(x)|^2$$

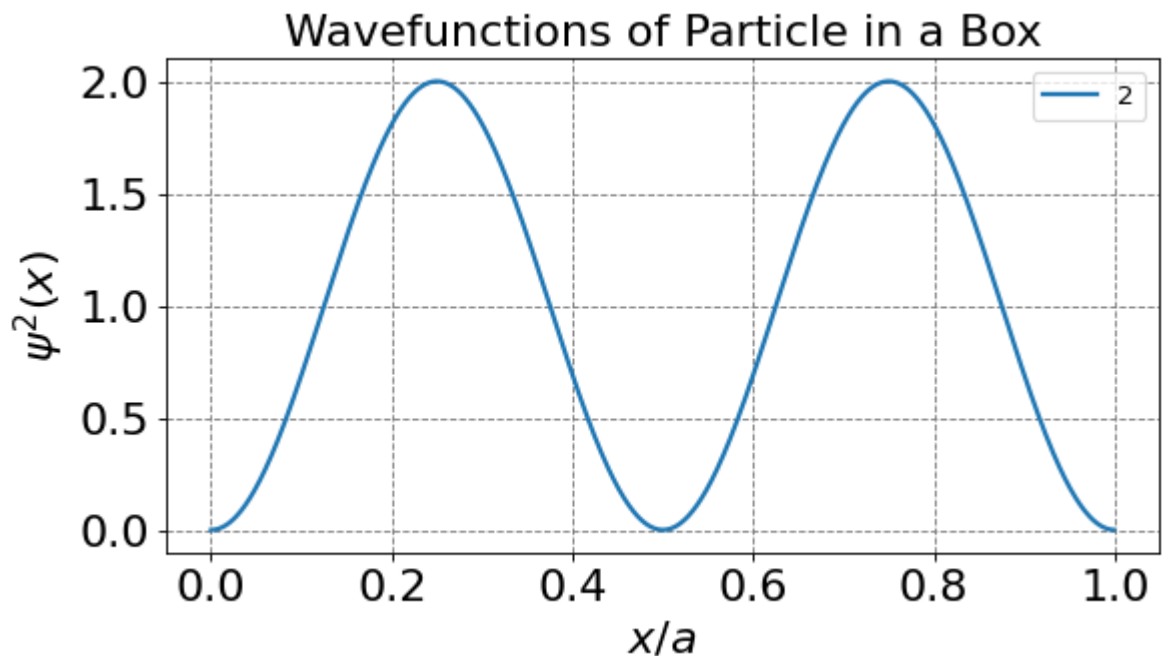
for **n = 1** and **n = 3** and select correct statements:

- ☐ Electron is equally likely everywhere
- ☐ Electron is never found at the walls
- ☐ Higher n leads to more oscillations
- ☐ Probability density can be negative

```
def psi_squared(x, n, a):
    return np.square(psi(x, n, a))
```

```
# make an array containing domain of wavelengths to consider
x = np.arange(0,1,0.001) # X is range
```

```
a=1 # L length of the box
n = 2 # Energy level
```

[Show code](#)


✓ Exercise 5: Nano-Connection (2–3 lines)

Pick ONE: Quantum dots / LEDs / Nanowires / Metal nanoparticles.

Prompt: Why does reducing size change color/electrical behavior?

Mini-Challenge (No math)

If an electron is confined in a **very tiny box**, its energy will be:

- ☐ Very small
- ☐ Moderate
- ☐ Very large

Explain in one sentence.

very large

✓ MCQ Quiz (Auto-graded, single attempt)

Complete the quiz in the next cell. Once submitted, answers will lock.

[Show code](#)

Particle in a Box Quiz

1) In a 1D particle-in-a-box, energy levels scale with quantum number n as:

- ☐ $E \propto n$
- ☒ $E \propto n^2$
- ☐ $E \propto 1/n$
- ☐ E is constant for all n

2) If the box length L increases, the energy spacing between levels:

- ☐ Increases
- ☒ Decreases
- ☐ Stays the same
- ☐ Becomes random

3) The number of nodes (zero crossings inside the box) in $\psi_n(x)$ is:

- ☐ n
- ☒ $n - 1$
- ☐ $2n$
- ☐ Always 0

4) The probability density $|\psi(x)|^2$:

- ☐ Can be negative
- ☒ Is always ≥ 0
- ☐ Is always constant
- ☐ Is undefined at the walls

5) Quantum confinement is most directly related to:

- ☐ Bigger objects having bigger mass
- ☐ Energy becoming continuous
- ☒ Discrete energies in nanoscale structures
- ☐ Classical friction in materials

Submit Answer

✓ Submitted! Score: 5/5



Report Generator (Final Submission)

Use the File -> Print to PDF option to generate your **PDF lab report** containing:

- Your details
- Completed tasks summary
- Key plots

Download and submit the PDF to your instructor.

